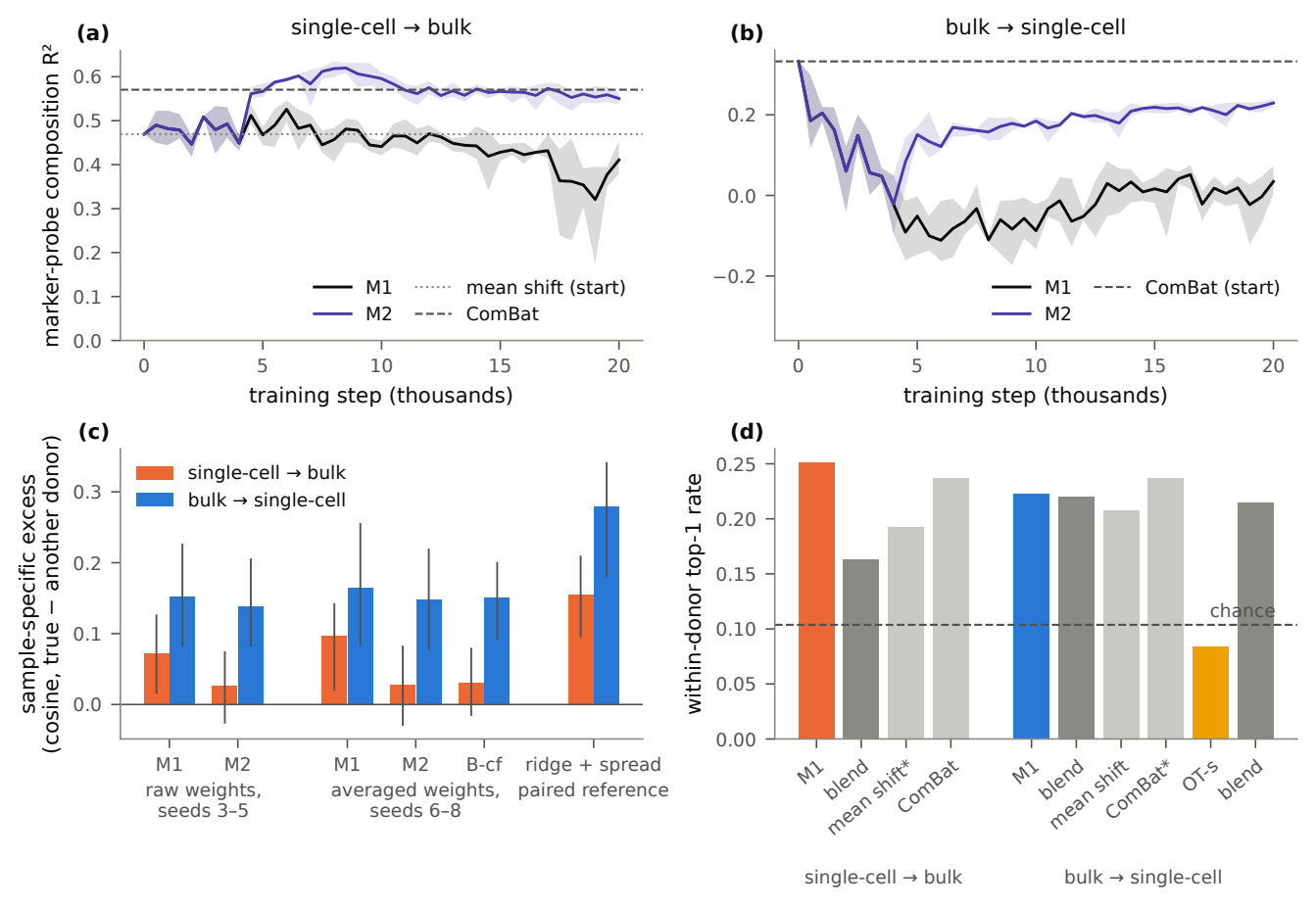
(a)
single-cell → bulk
marker-probe composition R²
training step (thousands)
M1
M2
mean shift (start)
ComBat
(b)
bulk → single-cell
training step (thousands)
M1
M2
ComBat (start)
(c)
single-cell → bulk
bulk → single-cell
sample-specific excess
(cosine, true − another donor)
M1
M2
raw weights,
seeds 3–5
M1
M2
B-cf
averaged weights,
seeds 6–8
ridge + spread
paired reference
(d)
within-donor top-1 rate
chance
M1
blend
mean shift*
ComBat
M1
blend
mean shift
ComBat*
OT-s
blend
single-cell → bulk
bulk → single-cell